UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	B	Which declaration creates an integer array of five elements?	int a[5]; reserves five integer elements.
A2	B	Given the declaration, what is displayed?	word[3] is 't' because array indexing starts at zero.
A3	B	Which operator reads data from cin?	The extraction operator >> reads from an input stream.
A4	A	What is displayed by the following code?	The loop displays 2, 4, 6 and 8 without spaces.
A5	D	Which of the following are valid identifiers?	total_score and _count are valid identifiers.
A6	C	What does a compiler do?	A compiler translates the complete source file into object code.
A7	A	Which declaration creates a floating-point variable named price?	float is used for real numbers.
A8	C	Which term describes a finite sequence of precise steps that solves a problem?	An algorithm is a finite, ordered set of unambiguous steps.
A9	C	Which of the following are selection structures?	if and switch select among alternatives.
A10	B	How many times does a while loop execute when its condition is false initially?	while is a pretest loop; a false condition skips the body.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: fundamentals

a) Define syntax error and logic error.

A syntax error violates the language grammar and is detected by the compiler. A logic error produces incorrect output although the program runs.

Marking: Award 1 mark for each definition.

(3 marks)

b) Explain TWO advantages of high-level languages over machine language.

They are easier to read and write, portable across machines, and supported by compilers and debugging tools.

Marking: Award 1.5 marks for each valid advantage.

(2 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the grade classification.

Start -> read score -> decision score >= 80 -> A; else score >= 60 -> B; else score >= 50 -> C; else score >= 40 -> D; else F -> display grade -> end.

Marking: Award marks for terminal symbols, input, four/five decisions, each output and correct flow.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

IF score >= 80 THEN grade = A ELSE IF score >= 60 THEN grade = B ... ENDIF; use if/else if in C++ and display grade.

Marking: Award marks for correct pseudocode, complete C++ structure, conditions, input and output.

(7 marks)

QUESTION 3

Topic: data types

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

Implicit conversion is automatic (int to double); explicit conversion uses a cast such as double(total) or static_cast<double>(total).

Marking: Award marks for both definitions and valid examples.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

double average = (m1 + m2 + m3) / 3.0; cout << fixed << setprecision(2) << average;

Marking: Award marks for floating-point division, fixed/setprecision and output.

(5 marks)

QUESTION 4

Topic: input output

a) Trace the program and state the output.

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Marking: The loop adds 1+2+3+4+5. Award marks for the sequence and final output.

(7 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

int i = 1; while (i <= 5) { total += i; i++; } With i < 5 the result is 10 because 5 is not added.

Marking: Award marks for correct while structure, update and changed result.

(8 marks)

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QUESTION 5

Topic: arrays

a) Declare an array named prices using the values in Table 1. Display all values.

double prices[5] = {20.5, 55.0, 15.9, 75.0, 10.0}; for (int i=0;i<5;i++) cout << prices[i] << endl;

Marking: Award marks for correct type, initialiser, loop and output.

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

double sale[5]; for(int i=0;i<5;i++){ sale[i] = prices[i] > 50 ? prices[i]*0.9 : prices[i]; } then display both arrays.

Marking: Award marks for second array, condition, discount, preservation and output.

(8 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A bookstore records up to five books. Table 1 shows an example of the fields.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

QUESTION 2

A catering company charges the meal prices shown in Table 1.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    string name;
    int quantity, grandQuantity = 0;
    double unitPrice, total, grandTotal = 0;
    char again;
    do {
        cout << "Customer name: ";
        getline(cin >> ws, name);
        cout << "Quantity: ";
        cin >> quantity;
        if (quantity < 0) {
            cout << "Invalid quantity" << endl;
        } else {
            if (quantity >= 200) unitPrice = 9.50;
            else if (quantity >= 100) unitPrice = 10.50;
            else if (quantity >= 50) unitPrice = 11.50;
            else unitPrice = 12.50;
            total = quantity * unitPrice;
```

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```
            grandTotal += total;
            grandQuantity += quantity;
            cout << fixed << setprecision(2);
            cout << "Receipt for " << name << ": "
                 << quantity << " x RM" << unitPrice
                 << " = RM" << total << endl;
        }
        cout << "Another customer? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    cout << "Total quantity: " << grandQuantity << endl;
    cout << "Total sales: RM" << fixed << setprecision(2)
         << grandTotal << endl;
    return 0;
}
```

Marking: Award 3 marks for input/validation, 6 for price selection and total, 3 for repetition, 2 for final totals and 1 for output clarity.

(15 marks)

END OF MARKING SCHEME

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